REVIEW 1 — ONE-PAGE SUMMARY

Temperature-Compensated PLL Frequency Synthesizer in sky130

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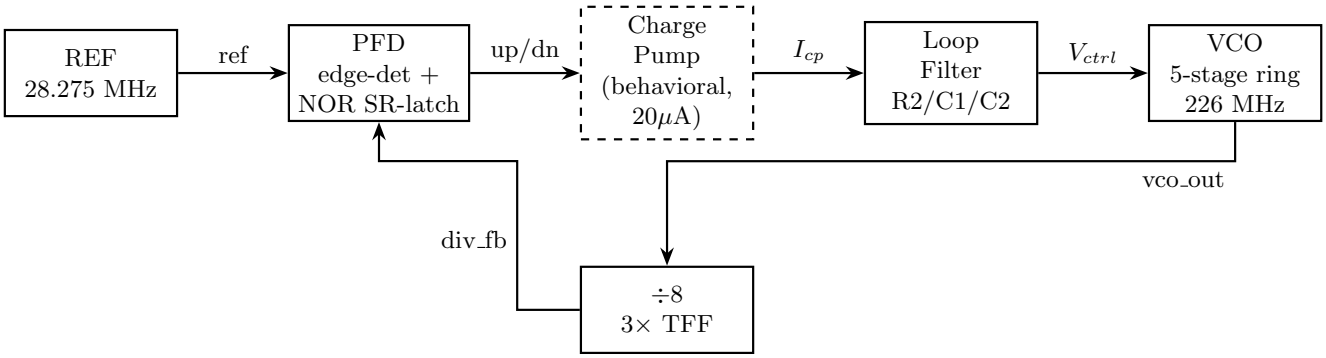
1 What changed and why

No local Cadence license ⇒ rebuilt the flow open-source: hand-written SPICE, ngspice, SkyWater sky130 PDK, fully headless (no GUI). Added a full schematic-to-GDS scope (original brief was schematic-only) and reframed the contribution around a literature-checked research question rather than reproducing a known circuit.

2 The hypothesis

Temperature compensation is a *cancellation* problem: a correction network subtracts two large drift terms to leave a small residual, which amplifies simulator/model disagreement by roughly the compensation ratio C. **H1:** $\varepsilon_{\text{res}} \approx C \cdot \varepsilon_{\text{raw}}$, implying a trust ceiling on how deep a correction an open-source flow can be relied on to design. Tested via cross-validation against Cadence/Spectre (Review 3).

3 Circuit built (all transistor-level except the charge pump)



Closed-loop lock (tt, 27°C): **577 ns** (spec <2 μs, averaged loop model 573 ns)

4 Key results

Quantity	Measured	Target / Reference
Closed-loop lock time	**577 ns**	$< 2\,\mu$s spec; 573 ns independent averaged model
Settled VCO / divider frequency	227.96 / 28.515 MHz	226.2 / 28.275 MHz target
K_{VCO} (local, @ 0.9V)	≈890 MHz/V	nonlinear curve; corrected from 670 (see erratum)
Uncompensated tempco, ff / tt / ss	332 / 681 / 1070 ppm/°C	> 3× spread across corners
Divider ratio (measured)	8.004	target 8.000

5 What we proved wrong (our own inherited spec, not the reference papers)

- **"±0.15 ppm stability"** is not physically meaningful for a reference-locked PLL ($f_{out} = N f_{ref}$ exactly in lock) → restated as $< \pm 0.3\%$ **residual VCO drift**, grounded against Souliotis et al. 2022's fabricated $-1.5\%/+0.4\%$.
- V_{ctrl} **floor of 0.27 V** is unreachable — this ring-VCO is device-threshold-limited, VCO is < 18 MHz below ≈0.65 V (no hard cutoff) → spec revised to **0.63 V**.
- A single scalar compensation ratio C doesn't hold given the > 3× corner spread → C **locked as per-corner**.

6 What we took from the reference papers, and what we added

- **Souliotis et al. 2022**: the PTAT/CTAT-compensated ring-VCO topology class (not claiming novelty here).
- **Freitas et al. 2025 (TWEPP)**: closest prior open-vs-commercial comparison, but theirs is *open-loop and qualitative only* → we close the loop and (Review 3) make it quantitative.
- **Thiriloganathan et al. 2026 (SMACD)**: confirms open-source tools have no PSS/Pnoise → we plan the same transient/jitter workaround, honestly labelled as an estimate.

Bottom line: the loop locks, in budget, with the measured lock time matching an independent averaged behavioural model (573 ns vs. 577 ns), and every specification gap found was caught with real simulation data, not assumed. Full detail, all raw numbers, and two documented root-cause debugging stories: `RESULTS.md` and `review1_report.pdf` in the project repository.

Erratum (2026-10-11). As first issued this summary quoted a "450 ns analytical prediction" and K_{VCO} ≈670 MHz/V (and the loop filter was reported as f_n ≈2 MHz, ζ ≈0.72, PM 70–72°). The loop-filter formula had dropped a $1/2\pi$; correct values are f_n ≈0.9 MHz, ζ ≈0.33, PM ≈30–33° (underdamped), and the local K_{VCO} at 0.9 V is ≈890 MHz/V. Simulated results (577 ns lock) are unchanged; R_2 retune is a Review 3 item. See `docs/loop_model.md`.